Heterogeneity, Selection, and Unemployment Dynamics Across Education Groups

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Abstract

Standard search-and-matching models predict that more productive workers have both lower separation rates and higher job-finding rates. However, in the data, more educated workers – who are more productive on average – have lower job-finding rates. I document that more educated workers are also substantially more heterogeneous in their wages and employment transitions. Motivated by these facts, I extend a standard search-and-matching model with endogenous separations to account for permanent productivity heterogeneity within education groups and calibrate it to match observed differences in wage levels and dispersion. In the model, more productive workers have higher job-finding rates and lower separation rates, but selection implies that less productive workers are overrepresented among the unemployed. As a result, measured job-finding rates can be lower for more educated workers despite higher individual job-finding probabilities. The calibrated model accounts for observed differences in unemployment rates, transition probabilities, and employment fluctuations across education groups.

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1. Introduction

Workers with higher levels of education experience substantially lower unemployment rates and more stable employment over the business cycle. A large empirical literature has documented these differences and attributed them primarily to lower separation rates among more educated workers. At the same time, an apparent puzzle emerges in the data: more educated workers do not exhibit higher job-finding rates. In fact, measured job-finding rates are often lower and more volatile for more educated workers. Standard search-and-matching models struggle to reconcile these patterns. In such models, more productive workers generate higher match surplus and should therefore experience both lower separation rates and higher job-finding rates. As a result, existing theoretical explanations have largely focused on mechanisms that equalize match surplus across education groups, such as differences in training requirements or hiring costs.

This paper provides a new explanation for these empirical patterns based on heterogeneity and selection within education groups. I document that more educated workers are not only more productive on average, but also substantially more heterogeneous in their wages and employment transitions. This heterogeneity generates systematic selection into employment and unemployment: more productive workers are more likely to be employed, while less productive workers are disproportionately represented among the unemployed. As a result, estimated transition probabilities do not reflect the probabilities faced by the average worker within each education group. In particular, measured job-finding rates are downward biased relative to the average individual job-finding probability, and this bias is larger for more educated workers because of their greater heterogeneity.

To formalize this mechanism, I develop a version of the Diamond–Mortensen–Pissarides model with endogenous job destruction and segmented labour markets. The model closely follows Mortensen and Pissarides (1994) and Cairó and Cajner (2018), but introduces permanent productivity heterogeneity within education groups. This extension preserves the standard surplus-based mechanism of search models: more productive workers generate higher match surplus, which increases vacancy creation, raises job-finding probabilities, and reduces separation rates. At the individual level, more productive workers in the model have both higher job-finding rates and lower separation rates. However, equilibrium selection implies that less productive workers are overrepresented among the unemployed, especially in more heterogeneous education groups. As a consequence, aggregate job-finding rates can be lower for more educated workers even though individual job-finding probabilities are higher on average.

I calibrate the model to match observed differences in wage levels and wage dispersion across education groups. The calibrated model successfully reproduces the key empirical patterns in unemployment dynamics across education groups, including lower unemployment rates, lower separation rates, and lower measured job-finding rates for more educated workers. The model also accounts for differences in employment volatility across education groups. These results demonstrate that standard surplus-based mechanisms can explain observed unemployment dynamics once heterogeneity and selection are properly accounted for.

This paper makes two main contributions. First, it provides new empirical evidence documenting substantial heterogeneity in wages and employment transitions within education groups, and shows that this heterogeneity is systematically larger among more educated workers. Second, it shows that incorporating this heterogeneity into an otherwise standard search-and-matching framework resolves the apparent inconsistency between theory and data. Contrary to existing explanations, the model does not require additional mechanisms that reduce the relative surplus of more educated workers, such as higher training time or hiring costs. Instead, differences in productivity and heterogeneity alone can account for the observed patterns.

The findings of this paper contribute to a large literature studying heterogeneity in unemployment dynamics. Previous work has emphasized differences in training requirements, hiring costs, or match-specific capital as explanations for cross-sectional differences in unemployment. This paper highlights a complementary and previously overlooked mechanism: selection arising from heterogeneity within education groups. More broadly, the results underscore the importance of accounting for composition effects when interpreting observed transition probabilities and unemployment dynamics.

1.1 Related literature

This paper contributes to the large literature studying unemployment dynamics using search-and-matching models. The theoretical framework builds on the seminal work of Mortensen and Pissarides (1994), who develop a model with endogenous job destruction and equilibrium unemployment fluctuations. Their framework predicts that more productive workers generate higher match surplus, leading to lower separation rates and higher job-finding rates.

This paper is closely related to Cairó and Cajner (2018), who study differences in unemployment dynamics across education groups using a search-and-matching model with endogenous separations and on-the-job training. They show that higher training requirements increase match surplus and reduce separation rates, while leaving job-finding rates largely unchanged. In contrast, this paper emphasizes permanent productivity heterogeneity within education groups and shows that selection arising from heterogeneity can generate lower measured job-finding rates for more educated workers, even when individual job-finding probabilities are higher.

The empirical motivation for this paper is closely related to work documenting differences in unemployment flows across education groups. Elsby et al. (2010) and Forsythe and Wu (2021) provide detailed empirical estimates of job-finding and separation rates by education, showing that more educated workers experience substantially lower separation rates but similar or lower job-finding rates. Earlier work by Mincer (1991) attributes lower separation rates among more educated workers to greater investments in job training and match-specific human capital.

This paper also relates to recent theoretical work proposing alternative explanations for differences in unemployment dynamics across workers. Sengul (2017) develops a model with information frictions in which firms learn about worker productivity over time, generating differences in separation and job-finding rates across workers. In contrast, this paper shows that permanent productivity heterogeneity and equilibrium selection alone can account for observed differences in unemployment dynamics across education groups within an otherwise standard search-and-matching framework.

Outline. The remainder of the paper is organized as follows. Section 2 documents key empirical patterns in unemployment dynamics across education groups. Section 3 presents the model. Section 4 describes the calibration strategy and parameter values. Section 5 presents the main results. Section 6 concludes.

2. Data

This section documents key empirical patterns in labor market dynamics across education groups. I combine three complementary data sources. First, I use monthly microdata from the Current Population Survey (CPS) to construct unemployment rates, employment rates, and transition probabilities for individuals 25 years of age and older. Transition probabilities are estimated following the methodology of Shimer (2012), which uses information on short-term unemployment to recover job-finding and separation rates from aggregate stocks. This approach avoids measurement error associated with directly observed gross flows and provides reliable estimates of transition rates at business-cycle frequencies. Appendix A shows that estimates constructed using CPS gross worker flows yield similar results.

Second, I use wage data from the CPS Annual Social and Economic Supplement (CPS-ASEC) to document differences in wage levels and dispersion across education groups. These data provide direct evidence on cross-sectional differences in earnings and productivity heterogeneity.

Third, I use longitudinal microdata from the National Longitudinal Survey of Youth 1997

(NLSY97) to study the distribution of individual employment spell durations. These panel data allow me to measure heterogeneity in employment stability within education groups and to characterize the dispersion of employment outcomes at the individual level.

Taken together, these datasets provide complementary evidence on transition rates, wage dispersion, and employment dynamics across education groups.

2.1 Unemployment volatility by education

Figure 1 plots the unemployment rate and employment-to-population ratio by education group. Less educated workers experience both higher unemployment rates and substantially larger fluctuations over the business cycle. In contrast, college-educated workers face significantly more stable labor market outcomes.

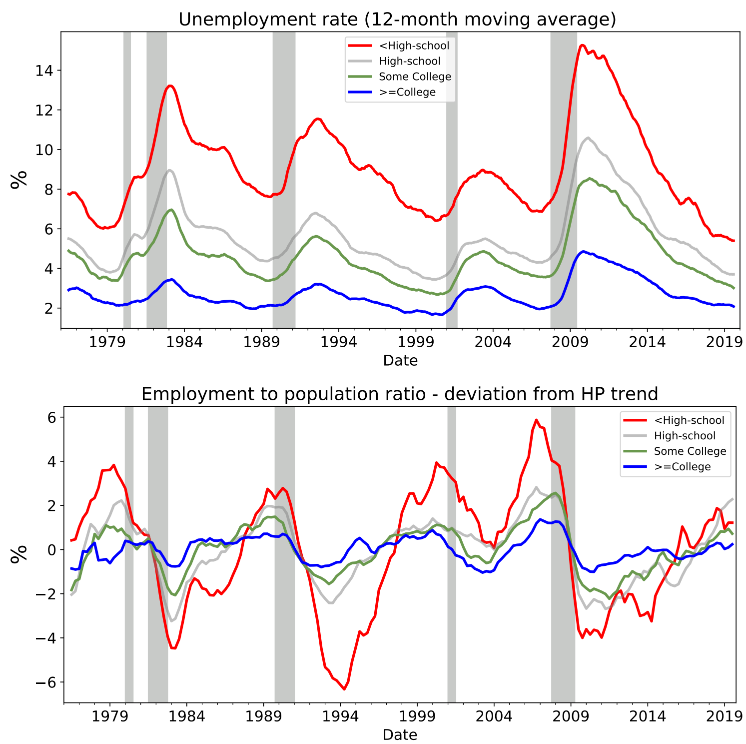

Figure 1: *Top:* Unemployment rate, by education level. *Bottom:* Employment-to-population ratio. Quarterly averages of seasonally-adjusted monthly CPS data for individuals 25 years of age and older.

Table 1 quantifies these differences. The volatility of the unemployment rate for individuals without a high school diploma is more than three times larger than for college graduates. Similar patterns hold for employment rates and employment-population ratios. These facts imply that aggregate labor market fluctuations disproportionately affect less educated workers.

Table 1: Labor market volatility by education level

	Unemployment rate (abs. deviation from trend)	Employment rate (log deviation from trend)	Employment-population ratio (log deviation from trend)
Less than high school	1.98	2.21	2.98
High school	1.40	1.50	1.60
Some college	1.15	1.22	1.20
College degree	0.61	0.62	0.72
Ratio (Low / College)	3.28	3.54	4.12

Quarterly averages of seasonally-adjusted CPS data, 1976–2019. Volatilities are defined as standard deviations of deviations from HP trends with smoothing parameter 10^5.

2.2 Transition probabilities

To understand the sources of these differences, Figure 2 plots job-finding and separation rates by education level. Separation rates differ substantially across education groups, with less educated workers facing significantly higher separation risk. In contrast, job-finding rates are much more similar across education groups and are slightly higher on average for less educated workers.

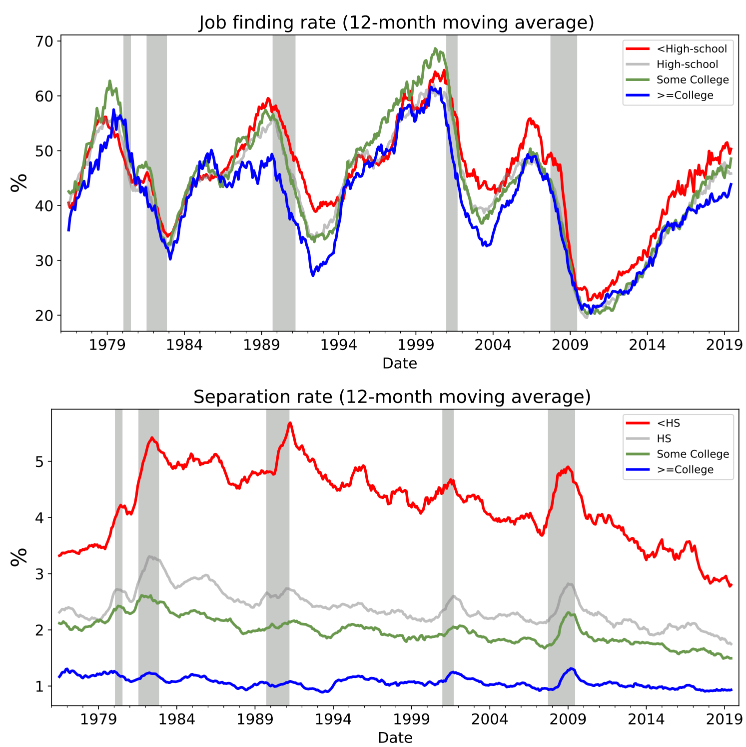

Figure 2: *Top:* Job-finding rate, by education level. *Bottom:* Separation rate, by education level. Transition probabilities are estimated following Shimer (2012).

These findings imply that differences in unemployment rates across education groups are driven primarily by differences in separation rates rather than job-finding rates. Appendix 1 shows that estimates based directly on CPS gross worker flows lead to similar conclusions.

2.3 Within-group heterogeneity

In addition to differences in average transition rates, education groups differ substantially in the dispersion of wages and employment outcomes. Table 2 reports mean wages and their standard deviations by education group using CPS data. Both the level and the dispersion of wages increase monotonically with education. College graduates earn on average 3.55 times more than workers without a high school diploma, while the standard deviation of wages is more than four times larger. This pattern indicates that more educated workers face substantially greater heterogeneity in productivity and match quality.

Table 2: Wage Moments by Education Group

	Mean Wage		Standard Deviation	
Education Group	Level	Relative to <HS	Level	Relative to <HS
<HS	24,906.8	1.00	1,222.3	1.00
HS	35,131.3	1.50	2,193.8	1.71
Some College	47,380.6	1.90	3,464.3	2.83
College	88,495.3	3.55	4,920.3	4.03

Notes: Real wages for CPS prime-aged males (2010–2019), expressed in constant dollars. Relative values are normalized to the least educated group (<HS).

Consistent with this evidence, panel data from the National Longitudinal Survey of Youth 1997 (NLSY97) reveal substantial differences in the dispersion of employment outcomes across education groups. Figure 3 plots the cross-sectional distribution of average employment spell duration. The dispersion of employment spells increases monotonically with education. More educated workers exhibit both very short and very long employment relationships, indicating a wider distribution of match quality and job stability.

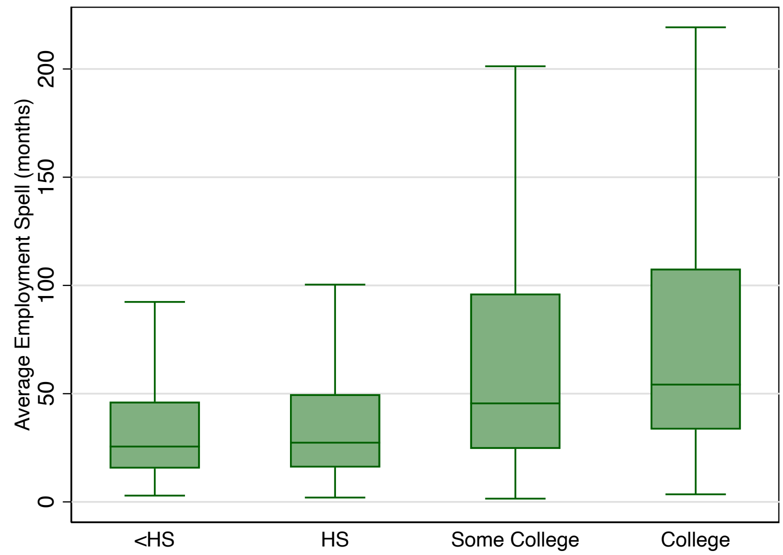

Figure 3: Employment spell duration by education (prime-age men, NLSY97)

Notes: Box-and-whisker plots show the distribution of average employment spell duration (months). Dispersion increases with education, indicating greater heterogeneity among more educated workers.

Taken together, these facts suggest that more educated workers face substantially greater heterogeneity in productivity and match quality. This heterogeneity is central to the mechanism proposed in this paper. In particular, greater dispersion implies stronger selection effects in employment and unemployment, which shape average transition probabilities and the cyclical behavior of unemployment across education groups.

3. Model

The empirical evidence presented in the previous section highlights three key facts. First, unemployment and employment are substantially more volatile for less educated workers. Second, these differences are driven primarily by higher separation rates rather than lower job-finding rates. Third, more educated workers exhibit significantly greater heterogeneity in wages and employment stability, indicating a wider dispersion in productivity and match quality.

Motivated by these facts, I develop a search and matching model with heterogeneous workers and segmented labor markets. I follow Cairó and Cajner (2018) and Mortensen and Pissarides (1994) in modeling an economy in which workers differ along two observable dimensions: education level and permanent productivity type. Education captures systematic differences across groups, while permanent productivity captures within-group heterogeneity.

Labor markets are segmented by worker type, reflecting the fact that education and permanent productivity are observable characteristics that allow firms to direct their recruiting efforts toward specific worker groups.

A central mechanism in the model is that greater productivity dispersion among more educated workers generates stronger selection effects in employment and unemployment. This mechanism plays a key role in shaping transition probabilities and aggregate labor market dynamics across education groups.

3.1 Environment

I consider a discrete-time economy populated by an infinite number of worker. Workers are heterogeneous in their education level and a permanent productivity type. Labour markets are indexed by pairs $(h, x) \in \mathcal{H} \times \mathcal{X}$, where h denotes the worker's education level and x denotes the worker's productivity. Segmented labour markets reflect the fact that education and permanent productivity are observable characteristics, allowing firms to direct their search toward specific worker types.

Each segmented labour market (h, x) features a continuum of measure one of infinitely-lived, risk-neutral workers and a large measure of risk-neutral firms. Workers maximize expected discounted lifetime utility

$$\mathbb{E}_t \sum_{k=0}^{\infty} \beta^k c_{t+k},$$

where $\beta \in (0, 1)$ is the discount factor and c_t denotes consumption.

3.2 Workers

Workers are characterized by their education level $H_h \in \mathbb{R}_+$ and permanent productivity type $x \in \mathcal{X} \subset \mathbb{R}_+$. Permanent productivity captures persistent differences in worker ability within education groups. Permanent productivity affects output multiplicatively.

At any point in time, workers can be either employed or unemployed. Employed workers earn wages, while unemployed workers receive flow payoff b, which captures home production, unemployment benefits. I abstract from labour force participation decisions, so all unemployed workers search for jobs.

3.3 Firms and production

Firms operate single-worker production units. A firm can post one vacancy at flow cost c, equal in all markets. Once matched with a worker of type (h, x), production is given by

$$y(H_h, x, A, a) = H_h \, x \, A \, a,$$

where:

- H_h denotes education-specific human capital,
- x denotes permanent worker productivity,
- A denotes aggregate productivity,
- a denotes match-specific idiosyncratic productivity.

Aggregate productivity evolves according to a Markov process

$$A' \sim P_A(A'|A).$$

Idiosyncratic productivity evolves according to an independent Markov process

$$a' \sim P_a(a'|a).$$

3.4 Matching

Matching in each segmented labour market (h, x) is governed by a constant-returns matching function

$$m(u, v) = \mu u^{\alpha} v^{1-\alpha},$$

where u and v denote unemployment and vacancies, $\mu > 0$ is matching efficiency, and $\alpha \in (0, 1)$.

Define labour market tightness

$$\theta(h, x, A) \equiv \frac{v(h, x, A)}{u(h, x, A)}.$$

The vacancy-filling probability is

$$q(\theta) = \mu \theta^{-\alpha}, \tag{1}$$

and the job-finding probability is

$$p(\theta) = \mu\theta^{1-\alpha}. \tag{2}$$

3.5 Firm value

Let $J(h, x, A, a)$ denote the value of a filled job. The Bellman equation is

$$J(h, x, A, a) = \max\left\{0,\ H_h xAa - w(h, x, A, a) + \beta(1-\delta)\mathbb{E}_{A',a'}[J(h, x, A', a')]\right\}. \tag{3}$$

The firm separates whenever match value is negative. Therefore, endogenous separations are characterized by reservation productivity $\tilde{a}(h, x, A)$ satisfying

$$J(h, x, A, \tilde{a}(h, x, A)) = 0. \tag{4}$$

Free entry implies

$$\frac{c}{q(\theta(h, x, A))} = \beta\mathbb{E}_{A',a'}[J(h, x, A', a')]. \tag{5}$$

3.6 Worker value

Let $W(h, x, A, a)$ denote the value of employment and $U(h, x, A)$ the value of unemployment.

The value of employment satisfies

$$W(h, x, A, a) = \max\left\{U(h, x, A),\ w(h, x, A, a) + \beta\delta\mathbb{E}_{A'}[U(h, x, A')] + \beta(1-\delta)\mathbb{E}_{A',a'}[W(h, x, A', a')]\right\}. \tag{6}$$

The value of unemployment satisfies

$$U(h, x, A) = b + \beta\mathbb{E}_{A'}\left[p(\theta(h, x, A))\mathbb{E}_{a'}[W(h, x, A', a')] + (1 - p(\theta(h, x, A)))U(h, x, A')\right]. \tag{7}$$

3.7 Wage determination

Wages are determined by generalized Nash bargaining. Let match surplus be

$$S(h, x, A, a) = J(h, x, A, a) + W(h, x, A, a) - U(h, x, A).$$

Surplus splitting implies

$$W(h, x, A, a) - U(h, x, A) = \gamma S(h, x, A, a),$$

$$J(h, x, A, a) = (1 - \gamma)S(h, x, A, a),$$

which implies wage equation

$$w(h, x, A, a) = \gamma [H_h x A a + c\theta(h, x, A)] + (1 - \gamma)b. \tag{8}$$

3.8 Equilibrium

A recursive equilibrium is a set of functions:

$$\{\theta(h, x, A), \tilde{a}(h, x, A), J(h, x, A, a), W(h, x, A, a), U(h, x, A), w(h, x, A, a)\}$$

such that, for all (h, x, A, a):

1. The firm value function satisfies equation (3).
2. Reservation productivity satisfies equation (4).
3. Free entry condition (5) holds.
4. Worker value functions satisfy equations (6) and (7).
5. Wages satisfy Nash bargaining condition (8).
6. Matching probabilities satisfy equations (1) and (2).

Employment and unemployment evolve according to endogenous separations and job finding probabilities in each segmented market (h, x).

3.9 Mechanism: Heterogeneity and Selection

This subsection formalizes how productivity heterogeneity generates differences between individual job-finding probabilities and measured job-finding rates at the group level.

Consider an education group h with permanent productivity distributed according to cumulative distribution function $F_h(x)$. Let $p_h(x, A)$ denote the individual job-finding probability for a worker with productivity x, which in equilibrium is given by

$$p_h(x, A) = \mu\theta(h, x, A)^{1-\alpha}.$$

Because more productive workers generate higher match surplus, free entry implies higher equilibrium labor market tightness $\theta(h, x, A)$ in markets with higher x. As a result, individual job-finding probabilities are increasing in productivity:

$$\frac{\partial p_h(x, A)}{\partial x} > 0.$$

Let $G_h^U(x, A)$ denote the productivity distribution among unemployed workers in education group h. The measured job-finding rate for group h is given by the average job-finding probability among unemployed workers:

$$f_h(A) = \int p_h(x, A)\, dG_h^U(x, A). \tag{9}$$

This object differs from the average job-finding probability across the entire population of workers in group h:

$$\bar{p}_h(A) = \int p_h(x, A)\, dF_h(x). \tag{10}$$

Selection arises because unemployment is not a random sample of workers. More productive workers have both higher job-finding probabilities and lower separation rates, making them more likely to remain employed. As a result, unemployed workers are negatively selected in terms of productivity. Formally, the productivity distribution of unemployed workers is first-order stochastically dominated by the population distribution:

$$G_h^U(x, A) \leq F_h(x) \quad \text{for all } x.$$

Since $p_h(x, A)$ is increasing in productivity, this implies

$$f_h(A) \leq \bar{p}_h(A). \tag{11}$$

The strength of this selection effect depends on the dispersion of productivity within the group. Education groups with greater productivity heterogeneity experience stronger selection into unemployment, because low-productivity workers are disproportionately represented among the unemployed. Consequently, measured job-finding rates can be lower for more educated workers even when their average individual job-finding probability is higher.

This mechanism reconciles the empirical finding that more educated workers have lower unemployment and separation rates, but similar or lower measured job-finding rates. The key driver is not lower individual job-finding probabilities, but stronger selection arising from greater within-group heterogeneity.

4. Calibration

This section describes the calibration of the model. The model is simulated at monthly frequency. Calibration proceeds in two steps. First, we assign parameter values that are common across all labour markets. Second, we calibrate productivity levels and heterogeneity across education groups to match empirical wage moments.

The key empirical targets are mean wages and log-wage dispersion across education groups. These moments are computed using microdata from the Current Population Survey (CPS) for prime-aged males (ages 25–54), between 2010 and 2019.

We divide workers into two education groups: (i) workers without a college degree ("No college"), which includes individuals with less than high school, high school, or some college education; and (ii) college graduates.

The model is solved by computing equilibrium labour market tightness and reservation productivity for each education group h and productivity type x, conditional on aggregate productivity A. The model is then simulated using the law of motion implied by equilibrium conditions.

4.1 Externally calibrated parameters

Table 3 reports parameter values that are common across all labour markets.

Table 3: Parameter Values at the Aggregate Level

Parameter	Interpretation	Value	Rationale
β	Discount factor	0.9966	Interest rate 4% p.a.
μ	Matching efficiency	0.334	Cairó and Cajner (2018) (CPS)
α	Matching function elasticity	0.5	Petrongolo and Pissarides (2001)
γ	Worker's bargaining power	0.5	Hosios condition
c	Vacancy posting cost	0.11	Cairó and Cajner (2018)
b	Value of being unemployed	0.71	Hall and Milgrom (2008)
σ_A	Std. dev. of log aggregate productivity	0.0058	Cairó and Cajner (2018), BLS
ρ_A	Persistence of log aggregate productivity	0.99	Cairó and Cajner (2018), BLS
λ	Probability of productivity change	1/3	Fujita and Ramey (2012)
δ	Exogenous separation rate	0.007	Cairó and Cajner (2018), JOLTS data

The discount factor β is set equal to 0.9966, corresponding to an annual real interest rate of 4 percent. The matching function is assumed Cobb–Douglas, with elasticity $\alpha = 0.5$, consistent with the empirical evidence surveyed by Petrongolo and Pissarides (2001). Matching efficiency μ is chosen to match the average job-finding rate in CPS data.

Worker bargaining power is set equal to $\gamma = 0.5$, satisfying the Hosios efficiency condition. The value of unemployment is set to $b = 0.71$, consistent with Hall and Milgrom (2008). Vacancy posting costs are set to $c = 0.11$, following Cairó and Cajner (2018).

The exogenous separation rate is set equal to $\delta = 0.007$, consistent with JOLTS evidence. Aggregate productivity follows an AR(1) process in logs with persistence $\rho_A = 0.99$ and innovation standard deviation $\sigma_A = 0.0058$, matching the volatility and persistence of U.S. labour productivity. Idiosyncratic productivity evolves stochastically with probability $\lambda = 1/3$, consistent with estimates in Fujita and Ramey (2012).

4.2 Internally calibrated parameters

The remaining parameters are calibrated jointly to match a set of aggregate labor market moments and cross-sectional wage moments in the data. The internally calibrated parameters are

$$\left\{b,\ \sigma_a,\ \mu_a,\ \sigma_{LS},\ \sigma_{HS},\ \frac{H_{HS}}{H_{LS}}\right\},$$

where b denotes the unemployment benefit (or outside option), σ_a and μ_a the standard deviation and mean of idiosyncratic match productivity shocks, σ_{LS} and σ_{HS} the dispersion of permanent productivity within each skill group, and H_{HS}/H_{LS} the relative human capital of high-skilled workers. H_{LS} is normalized to 1. Given σ_a, the mean idiosyncratic productivity μ_a is chosen to make $\mathbb{E}[a] = 1$, implying $\mu_a = 1 - \frac{\sigma_a^2}{2}$. So we are left with five parameters to calibrate jointly to match five moments in the data.

Targeted moments in the data. The calibration targets five empirical moments:

1. The mean aggregate job-finding rate, $\mathbb{E}[f]$.
2. The mean aggregate separation rate, $\mathbb{E}[s]$.
3. The standard deviation of log hourly wages for low-skilled workers, $\text{Std}(\log w_{LS})$.
4. The standard deviation of log hourly wages for high-skilled workers, $\text{Std}(\log w_{HS})$.
5. The ratio of mean hourly wages between high- and low-skilled workers,

$$\frac{\mathbb{E}[w_{HS}]}{\mathbb{E}[w_{LS}]}.$$

The labor market flow moments are computed from CPS data using standard methods, as described in the Data section. that recover job-finding and separation rates from unemployment stocks and short-term unemployment. Wage moments are computed from CPS hourly wage data. For each education group, the standard deviation of log wages is calculated across individuals, and mean wages are computed using the same sample restrictions.

Model counterparts. For any candidate parameter vector, the model is simulated for a long time horizon. Aggregate job-finding and separation rates are computed from simulated unemployment flows exactly as in the data. Wage moments are computed from simulated wages in the same way as in the CPS: within each skill group, we compute the cross-sectional standard deviation of log wages and the mean wage level. The wage ratio is then constructed directly from simulated group means.

Joint calibration. The five parameters are chosen simultaneously to minimize the distance between model-implied moments and their empirical counterparts. Formally, letting m^{data} denote the vector of targeted moments and $m^{\text{model}}(\theta)$ the corresponding model moments, the calibrated parameter vector θ solves

$$\min_{\theta} \; \|m^{\text{model}}(\theta) - m^{\text{data}}\|,$$

where $\theta = \{b, \sigma_a, \sigma_{LS}, \sigma_{HS}, H_{HS}/H_{LS}\}$. This joint calibration ensures that unemployment dynamics and wage inequality across and within skill groups are disciplined simultaneously in equilibrium.

Table 4: Internal Calibration: Parameters and Targeted Moments

Symbol	Interpretation	Value	Targeted Moment	Data	Model
b	Unemployment benefit / outside option	0.75	$\mathbb{E}[f]$	43.46	42.99
σ_a	Std. dev. of idiosyncratic productivity	0.21	$\mathbb{E}[s]$	2.19	2.22
σ_{LS}	Within-group dispersion (Low-Skilled)	0.11	$\text{Std}(\log w_{LS})$	0.35	0.27
σ_{HS}	Within-group dispersion (High-Skilled)	0.23	$\text{Std}(\log w_{HS})$	0.55	0.42
$\frac{H_{HS}}{H_{LS}}$	Relative human capital	2.56	$\frac{\mathbb{E}[w_{HS}]}{\mathbb{E}[w_{LS}]}$	2.41	2.40

Notes: Flow moments are expressed in percent. Wage moments are computed using CPS hourly wage data. Parameters are chosen jointly to minimize the distance between model and data moments.

5. Results

The calibrated model successfully reproduces key empirical differences in unemployment dynamics between college and non-college workers. In particular, the model generates lower unemployment rates, lower separation rates, and lower job-finding rates for college graduates, consistent with observed data. The model also accounts for differences in employment volatility across education groups.

We begin with the aggregate economy. As shown in Table 5, the mean job-finding and separation rates are closely matched by construction, as these are explicit calibration targets. The implied unemployment rate is not directly targeted, but is determined by the steady-state flow condition and is reproduced reasonably well by the model.

Table 5: Aggregate Moments: Data vs. Model (Percent)

	Unemployment rate (u)		Job finding rate (f)		Separation rate (s)	
Moment	Data	Model	Data	Model	Data	Model
Mean	4.76	5.06	43.36	42.99	2.19	2.22
Std. Dev.	1.16	0.74	8.14	3.91	0.29	0.17

Notes: All moments are expressed in percent. f denotes the job-finding rate and s the separation rate. Model moments are computed from a long simulation of the aggregate economy.

However, the model underpredicts aggregate volatility. The standard deviations of unemployment, job-finding, and separation rates are all lower than in the data. This feature is well known in the search-and-matching literature and closely related to the Shimer (2005) volatility puzzle: standard DMP-type models struggle to generate sufficient amplification of labor market fluctuations without additional propagation mechanisms. In this sense, the aggregate results are consistent with existing findings in the literature.

The main contribution of the model emerges when we examine education-specific economies. Tables 6 and 7 report the results separately for high- and low-skilled workers. Several important patterns arise.

First, the model generates lower unemployment rates for high-skilled workers. Although the quantitative gap is smaller than in the data, the ranking across groups is correctly reproduced. Second, the model generates lower separation rates for high-skilled workers, consistent with the empirical evidence that more educated workers experience more stable employment relationships.

Table 6: High-Skilled: Data vs. Model (Percent)

	Unemployment rate (u)		Job finding rate (f)		Separation rate (s)	
Moment	Data	Model	Data	Model	Data	Model
Mean	2.70	4.98	40.12	40.69	1.27	2.06
Std. Dev.	0.61	0.43	9.12	5.10	0.18	0.10

Notes: All moments are expressed in percent. f denotes the job-finding rate and s the separation rate. Data moments are computed from CPS microdata. Model moments are computed from simulated time series.

Table 7: Low-Skilled: Data vs. Model (Percent)

	Unemployment rate (u)		Job finding rate (f)		Separation rate (s)	
Moment	Data	Model	Data	Model	Data	Model
Mean	5.52	6.29	43.41	43.02	2.54	2.64
Std. Dev.	1.42	0.91	8.25	4.24	0.27	0.14

Notes: All moments are expressed in percent. f denotes the job-finding rate and s the separation rate. Data moments are computed from CPS microdata. Model moments are computed from simulated time series.

Most importantly, the model generates both a lower and more volatile job-finding rate for high-skilled workers. This is a key result. In standard search-and-matching models without selection, higher productivity types typically exhibit higher job-finding rates, since greater match surplus encourages vacancy creation. In contrast, in this model, high-skilled workers face stricter selection: firms are more selective when posting vacancies for more productive workers, which reduces the average job-finding probability while amplifying its cyclical sensitivity.

Thus, despite being more productive, high-skilled workers experience lower job-finding rates and greater job-finding volatility in the model. This mechanism—driven by endogenous selection— allows the model to generate patterns that standard homogeneous DMP frameworks cannot reproduce. The differential cyclicality of job-finding rates across education groups is therefore not a consequence of lower productivity, but rather of tighter acceptance thresholds and stronger composition effects in high-skilled labor markets.

5.1 Individual moments and selection

Figure 4 plots mean unemployment, job-finding, and separation rates as functions of individual permanent productivity, $y \equiv H_h x$, where H_h denotes education-specific human capital and x

the idiosyncratic productivity component.

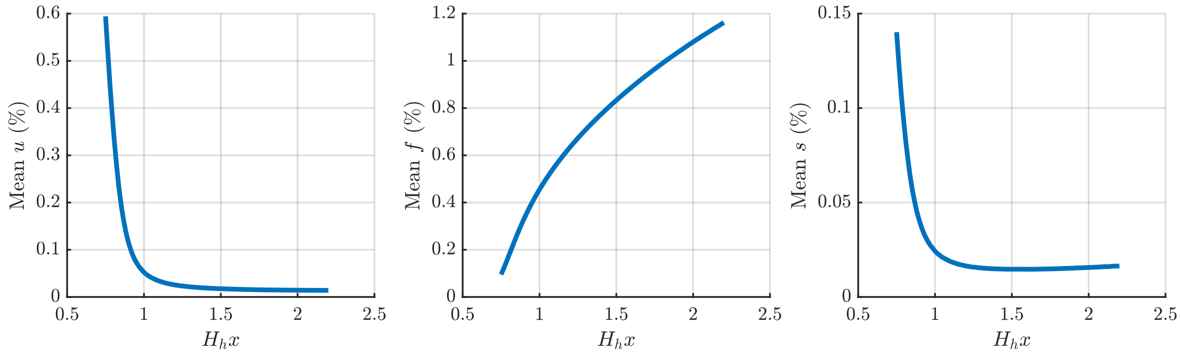

Figure 4: Labor market moments and permanent productivity. Mean unemployment, job-finding, and separation rates as functions of individual permanent productivity $H_h x$.

At the individual level, higher productivity increases job finding. For a worker of type y, the job-finding rate is

$$f(y) = \mu\,\theta(y)^{1-\alpha}, \tag{12}$$

where $\theta(y)$ denotes labor market tightness in the submarket directed toward workers of productivity y. Because higher y raises match surplus, equilibrium tightness $\theta(y)$ is increasing in productivity, implying $f'(y) > 0$. The steady-state unemployment rate satisfies the flow condition

$$u(y) = \frac{s(y)}{s(y) + f(y)}, \tag{13}$$

so that higher productivity — which increases $f(y)$ and weakly reduces separations $s(y)$ — leads to lower unemployment. Figure 4 therefore illustrates a monotonic relationship: more productive workers experience both higher job-finding rates and lower unemployment.

High-skilled workers are, on average, more productive and therefore lie further to the right in the productivity distribution. However, observed job-finding rates are not simple averages across individuals; they are averages over the unemployed pool. Aggregate job finding is given by

$$\bar{f} = \int f(y)\,\omega(y)\,dy, \qquad \omega(y) = \frac{u(y)g(y)}{\int u(\tilde{y})g(\tilde{y})d\tilde{y}}, \tag{14}$$

where $g(y)$ is the population density of productivity and $\omega(y)$ is the distribution of productivity among the unemployed.

Because unemployment $u(y)$ is higher for low-productivity workers, these types are disproportionately represented in the unemployment pool. Since they also have lower individual job-finding rates, the composition weight $\omega(y)$ mechanically tilts the observed average toward lower values. This endogenous selection effect implies that group-level job-finding rates need not mirror individual productivity rankings. Even though high-skilled workers are more productive at the individual level, their observed job-finding rate reflects the composition of their unemployment pool. The figure therefore clarifies how individual productivity gradients and equilibrium selection jointly shape aggregate and group-level labor market moments.

5.2 The role of heterogeneity and selection

To illustrate the role of heterogeneity and selection, we compare the calibrated model to a counterfactual version without productivity heterogeneity. In the counterfactual model, all workers within each education group have identical productivity equal to the group mean. This eliminates selection effects and isolates the role of within-group heterogeneity.

Table 8: Labor Market Moments: Role of Productivity Heterogeneity

	Unemployment rate (u)			Job finding rate (f)			Separation rate (s)		
Education group	Data	Model	No heterogeneity	Data	Model	No heterogeneity	Data	Model	No heterogeneity
No college	5.52	6.29	5.55	43.41	43.02	35.02	2.54	2.64	2.06
College graduates	2.70	4.98	2.82	40.12	40.69	46.48	1.27	2.06	1.35

Notes: Labor market transition rates (in percent) by education group. Data moments are computed from CPS microdata. The baseline model includes permanent productivity heterogeneity within education groups. The "No heterogeneity" model sets productivity dispersion to zero within each group ($\sigma_h(x) = 0$).

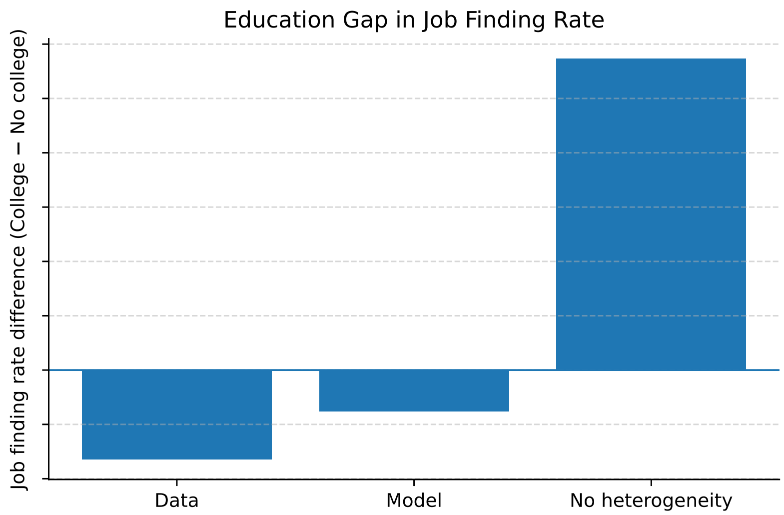

Figure 5: Education gap in job finding rates. This figure shows the difference in job finding rates between college graduates and workers without a college degree (College − No college), in the data and in the model. The baseline model reproduces the qualitative pattern observed in the data. In contrast, the counterfactual model without productivity heterogeneity generates a substantially different gap, highlighting the importance of within-group heterogeneity and selection. The vertical axis measures the difference in job finding rates, and the horizontal line denotes zero difference. Numerical values are omitted for confidentiality.

6. Conclusion

This paper argues that within-group heterogeneity and equilibrium selection are central to understanding unemployment dynamics across education groups. While standard search-and-matching models imply that more productive workers should both separate less and find jobs more quickly, the data show that college graduates have lower separation rates but similar or lower measured job-finding rates. I document that wage and employment outcomes are substantially more dispersed among more educated workers, and I show that this dispersion generates stronger selection into unemployment: low-productivity types are disproportionately represented among the unemployed, especially in the college group.

A calibrated Mortensen–Pissarides model with endogenous job destruction and permanent productivity heterogeneity reproduces the joint pattern of lower unemployment, lower separations, and lower measured job-finding rates for college graduates. In contrast, a counterfactual version without within-group heterogeneity fails to match the observed education gap in job-finding rates, underscoring the quantitative importance of composition effects. These findings suggest that observed transition probabilities should be interpreted with caution: differences across groups may reflect selection as much as differences in underlying job-finding opportunities. More broadly, the results highlight the value of incorporating heterogeneity within observable groups when using search models to interpret labor market flows and the distributional incidence of business-cycle fluctuations.

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